

MATH 2111

Tutorial 2: Matrix Algebra, Inverses and Determinants

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Matrix Products, Transposes and Inverses

Matrix products

If $A \in \mathbb{R}^{m \times n}$ and $B = [\mathbf{b}_1 \ \cdots \ \mathbf{b}_p] \in \mathbb{R}^{n \times p}$, then

$$AB = [A\mathbf{b}_1 \ \cdots \ A\mathbf{b}_p] \in \mathbb{R}^{m \times p}.$$

Matrix multiplication is associative and distributive, but in general

$$AB \neq BA, \quad AB = AC \not\Rightarrow B = C,$$

$$AB = 0 \not\Rightarrow A = 0 \text{ or } B = 0.$$

Order-sensitive identities

For products of compatible sizes,

$$(AB)^T = B^T A^T.$$

If A and B are invertible square matrices, then

$$(AB)^{-1} = B^{-1}A^{-1}, \quad (A^T)^{-1} = (A^{-1})^T.$$

The order of the factors is reversed in both identities.

Invertibility and Determinant Techniques

Selected equivalent conditions

For an $n \times n$ matrix A , the following are equivalent:

- A is invertible;
- A is row equivalent to I_n ;
- A has n pivot positions;
- $Ax = 0$ has only the trivial solution;
- the columns of A are linearly independent;
- the columns of A span $\mathbb{R}^n$;
- $x \mapsto Ax$ is one-to-one and onto.

Moreover,

$$[A \mid I_n] \sim [I_n \mid A^{-1}]$$

whenever A is invertible.

Determinant calculations

For $C_{ij} = (-1)^{i+j} \det A_{ij}$,

$$\det A = \sum_{j=1}^n a_{ij} C_{ij} \quad (\text{fixed row } i),$$

$$\det A = \sum_{i=1}^n a_{ij} C_{ij} \quad (\text{fixed column } j).$$

A triangular matrix has determinant equal to the product of its diagonal entries. Under row operations,

operation	effect on $\det$
$R_j \leftarrow R_j + cR_i$	unchanged
$R_i \leftrightarrow R_j$	multiplied by -1
$R_i \leftarrow cR_i$ ($c \neq 0$)	multiplied by c

Problem 1: Null Directions and Zero Products

Problem 1

Let

$$A = \begin{bmatrix} 2 & -8 \\ -1 & 4 \end{bmatrix}.$$

- (a) Find all solutions of $A\mathbf{x} = \mathbf{0}$.
- (b) Construct a 2×2 matrix B whose two columns are nonzero and distinct, such that $AB = \mathbf{0}$.
- (c) Use the column interpretation of matrix multiplication to explain the construction.
- (d) State at least four conclusions about A from the Invertible Matrix Theorem.

Solution to Problem 1

Since the second row is $-\frac{1}{2}$ times the first, the equation $A\mathbf{x} = 0$ reduces to

$$2x_1 - 8x_2 = 0 \iff x_1 = 4x_2.$$

Hence

$$\mathbf{x} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

Let $\mathbf{v} = (4, 1)^T$. Choosing two distinct nonzero multiples of $\mathbf{v}$ as the columns gives

$$B = \begin{bmatrix} 4 & 8 \\ 1 & 2 \end{bmatrix} = [\mathbf{v} \ 2\mathbf{v}].$$

Therefore

$$AB = [A\mathbf{v} \ A(2\mathbf{v})] = [\mathbf{0} \ \mathbf{0}].$$

Both columns of B lie in the null space of A , so the product vanishes even though neither factor is the zero matrix.

Solution to Problem 1: Consequences

Since $Ax = 0$ has a nontrivial solution, the Invertible Matrix Theorem gives:

- A is **not invertible**;
- A is not row equivalent to I_2 and has fewer than two pivot positions;
- the columns of A are linearly dependent and do not span $\mathbb{R}^2$;
- the map $x \mapsto Ax$ is neither one-to-one nor onto $\mathbb{R}^2$;
- no 2×2 matrix C can satisfy either $CA = I_2$ or $AC = I_2$.

Cancellation and zero products

The equality $AB = 0$ does not force $A = 0$ or $B = 0$. For a square matrix A , cancellation from $AB = AC$ for all compatible B, C is valid precisely when A is invertible.

Problem 2: An Order-Sensitive Matrix Equation

Problem 2

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 5 \\ -2 & 2 \end{bmatrix}.$$

Without introducing four scalar unknowns for the entries of X :

- (a) solve the matrix equation $AXB^T = C$;
- (b) verify the result by direct substitution;
- (c) show that transposing the equation gives $C^T = BX^T A^T$, and prove that the solution is unique.

Solution to Problem 2: Solving in the Correct Order

Both A and B are invertible, with

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \quad B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 \\ -\frac{1}{2} & 1 \end{bmatrix}.$$

Left-multiply by A^{-1} and right-multiply by $(B^T)^{-1}$:

$$AXB^T = C \implies X = A^{-1}C(B^T)^{-1}.$$

Since $(B^T)^{-1} = (B^{-1})^T$,

$$\begin{aligned} X &= \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 3 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 \end{bmatrix} = \boxed{\begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}}. \end{aligned}$$

Solution to Problem 2: Verification and Uniqueness

Direct substitution gives

$$AX = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix}, \quad AXB^T = \begin{bmatrix} 0 & 5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ -2 & 2 \end{bmatrix} = C.$$

Taking transposes reverses the order:

$$C^T = (AXB^T)^T = (B^T)^T X^T A^T = BX^T A^T.$$

If X_1 and X_2 both satisfy the equation, then

$$A(X_1 - X_2)B^T = 0.$$

Multiplication by A^{-1} on the left and $(B^T)^{-1}$ on the right gives $X_1 - X_2 = 0$. Hence the displayed matrix is the unique solution.

Problem 3: One Reduction, Several Right-Hand Sides

Problem 3

Let

$$A = \begin{bmatrix} -2 & -7 & -9 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}, \quad \mathbf{b}_1 = \begin{bmatrix} -7 \\ 6 \\ 3 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -11 \\ 7 \\ 5 \end{bmatrix}.$$

Use a **single** row reduction of

$$[A \mid \mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{e}_3]$$

to solve $A\mathbf{x} = \mathbf{b}_1$, solve $A\mathbf{x} = \mathbf{b}_2$, and find the third column of A^{-1} without computing its other columns. Explain why the same row operations solve all three tasks.

Solution to Problem 3: Forward Elimination

Start with

$$\left[\begin{array}{ccc|ccc} -2 & -7 & -9 & -7 & -11 & 0 \\ 2 & 5 & 6 & 6 & 7 & 0 \\ 1 & 3 & 4 & 3 & 5 & 1 \end{array} \right].$$

After $R_1 \leftrightarrow R_3$, eliminate the first column:

$$\xrightarrow{\substack{R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 + 2R_1}} \left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 3 & 5 & 1 \\ 0 & -1 & -2 & 0 & -3 & -2 \\ 0 & -1 & -1 & -1 & -1 & 2 \end{array} \right].$$

Then $R_2 \leftarrow -R_2$ and $R_3 \leftarrow R_3 + R_2$ give

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 3 & 5 & 1 \\ 0 & 1 & 2 & 0 & 3 & 2 \\ 0 & 0 & 1 & -1 & 2 & 4 \end{array} \right].$$

Solution to Problem 3: Backward Elimination

Continue from the echelon form:

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 3 & 5 & 1 \\ 0 & 1 & 2 & 0 & 3 & 2 \\ 0 & 0 & 1 & -1 & 2 & 4 \end{array} \right].$$

Applying

$$R_2 \leftarrow R_2 - 2R_3, \quad R_1 \leftarrow R_1 - 4R_3, \quad R_1 \leftarrow R_1 - 3R_2$$

yields

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 1 & 0 & 2 & -1 & -6 \\ 0 & 0 & 1 & -1 & 2 & 4 \end{array} \right].$$

Therefore

$$\boxed{\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}}, \quad \boxed{\mathbf{x}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}}, \quad \boxed{\text{col}_3(A^{-1}) = \begin{bmatrix} 3 \\ -6 \\ 4 \end{bmatrix}}.$$

Solution to Problem 3: Why One Reduction Suffices

Let E denote the product of the elementary matrices representing the row operations. The left block is reduced to the identity, so

$$EA = I_3.$$

Since A is square, this implies $E = A^{-1}$. The same operations act on every appended column:

$$E[\mathbf{b}_1 \ \mathbf{b}_2 \ \mathbf{e}_3] = [A^{-1}\mathbf{b}_1 \ A^{-1}\mathbf{b}_2 \ A^{-1}\mathbf{e}_3].$$

Hence the first two transformed columns are the solutions of the two systems, while

$$A^{-1}\mathbf{e}_3 = \text{col}_3(A^{-1}).$$

A direct check is

$$A\mathbf{x}_1 = \mathbf{b}_1, \quad A\mathbf{x}_2 = \mathbf{b}_2, \quad A \begin{bmatrix} 3 \\ -6 \\ 4 \end{bmatrix} = \mathbf{e}_3.$$

Problem 4: A Structured Family of Inverses

Problem 4

For $n \geq 2$, let

$$L_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 1 & 1 & 0 & \cdots & 0 \\ 1 & 1 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{bmatrix}.$$

- (a) Find L_4^{-1} and use it to conjecture a formula for L_n^{-1} .
- (b) Prove the formula by verifying both $L_n L_n^{-1} = I_n$ and $L_n^{-1} L_n = I_n$.
- (c) Describe the operations represented by L_n and L_n^{-1} on a vector.

Solution to Problem 4: Pattern

Computing the inverse for $n = 4$ gives

$$L_4^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

This suggests the lower bidiagonal matrix $D_n = (d_{ij})$, where

$$d_{ij} = \begin{cases} 1, & i = j, \\ -1, & i = j + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently,

$$D_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ -1 & 1 & 0 & \cdots & 0 \\ 0 & -1 & 1 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & 1 \end{bmatrix}.$$

Solution to Problem 4: Verification

In $D_n L_n$, the first row equals the first row of L_n . For $i \geq 2$, row i is

$$(\text{row } i \text{ of } L_n) - (\text{row } i - 1 \text{ of } L_n) = e_i^T.$$

Thus $D_n L_n = I_n$.

For $j < n$, column j of D_n is $e_j - e_{j+1}$, while column n is e_n . Hence

$$L_n(e_j - e_{j+1}) = \text{col}_j(L_n) - \text{col}_{j+1}(L_n) = e_j,$$

and $L_n e_n = e_n$. Therefore $L_n D_n = I_n$.

Consequently,

$$\boxed{L_n^{-1} = D_n}.$$

Solution to Problem 4: Partial Sums and Differences

For $\mathbf{x} = (x_1, \dots, x_n)^T$,

$$L_n \mathbf{x} = \begin{bmatrix} x_1 \\ x_1 + x_2 \\ x_1 + x_2 + x_3 \\ \vdots \\ x_1 + \dots + x_n \end{bmatrix}.$$

Thus L_n forms successive partial sums.

For $\mathbf{y} = (y_1, \dots, y_n)^T$,

$$L_n^{-1} \mathbf{y} = D_n \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 - y_1 \\ y_3 - y_2 \\ \vdots \\ y_n - y_{n-1} \end{bmatrix}.$$

Thus L_n^{-1} recovers consecutive differences from the partial sums.

Problem 5: Determinants by Structure

Problem 5

Let

$$M = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 5 & 0 & 8 \\ -1 & 3 & 5 & 2 \\ 0 & 1 & 0 & 4 \end{bmatrix}.$$

- (a) Choose an efficient row or column for a cofactor expansion and compute $\det M$.
- (b) Recompute $\det M$ by using only row replacements to reduce M to upper triangular form.
- (c) A matrix N is obtained from M by first interchanging the first two rows and then multiplying the last row by -3 . Find $\det N$ without reducing N .

Solution to Problem 5: Cofactor Expansion

The third column has only one nonzero entry. Expanding down that column,

$$\det M = 5(-1)^{3+3} \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 0 & 1 & 4 \end{vmatrix}.$$

Use determinant-preserving row replacements in the 3×3 minor:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 4 \end{bmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix}.$$

The minor has determinant 2, so

$$\boxed{\det M = 5 \cdot 2 = 10}.$$

Solution to Problem 5: Triangularization and Row Effects

Apply the row replacements in the displayed order:

$$\begin{aligned} R_2 &\leftarrow R_2 - 2R_1, & R_3 &\leftarrow R_3 + R_1, \\ R_3 &\leftarrow R_3 - 5R_2, & R_4 &\leftarrow R_4 - R_2. \end{aligned}$$

This gives

$$U = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 5 & -5 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Since row replacements leave the determinant unchanged,

$$\det M = \det U = (1)(1)(5)(2) = 10.$$

For N , the row interchange contributes -1 , and scaling the last row by -3 contributes -3 . Hence

$$\boxed{\det N = (-1)(-3) \det M = 30}.$$

Problem 6: A Parametric Invertibility Problem

Problem 6

For $\lambda \in \mathbb{R}$, let

$$A_\lambda = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 6 & 1 & 2 \\ -1 & 4 & \lambda + 3 & 2 \\ 1 & 0 & 2\lambda - 1 & \lambda + 6 \end{bmatrix}, \quad T_\lambda(\mathbf{x}) = A_\lambda \mathbf{x}.$$

- (a) Use only row replacements to reduce A_λ to upper triangular form and compute $\det A_\lambda$.
- (b) Determine all λ for which A_λ is invertible, using the pivot criterion.
- (c) For each exceptional value of λ , find a nonzero solution of $A_\lambda \mathbf{x} = \mathbf{0}$.
- (d) Describe when T_λ is one-to-one and onto, and describe the solution behavior of $A_\lambda \mathbf{x} = \mathbf{b}$.

Solution to Problem 6: Parameter-Free Elimination

First apply

$$R_2 \leftarrow R_2 - 2R_1, \quad R_3 \leftarrow R_3 + R_1, \quad R_4 \leftarrow R_4 - R_1.$$

This gives

$$\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 6 & \lambda + 3 & 3 \\ 0 & -2 & 2\lambda - 1 & \lambda + 5 \end{bmatrix}.$$

Next apply

$$R_3 \leftarrow R_3 - 3R_2, \quad R_4 \leftarrow R_4 + R_2, \quad R_4 \leftarrow R_4 - 2R_3.$$

The result is

$$U_\lambda = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & \lambda & 3 \\ 0 & 0 & 0 & \lambda - 1 \end{bmatrix}.$$

No division by a parameter has been used, so this reduction is valid for every $\lambda \in \mathbb{R}$.

Solution to Problem 6: Determinant and Generic Case

Every operation used above is a row replacement, so the determinant is unchanged. Since U_λ is upper triangular,

$$\det A_\lambda = \det U_\lambda = (1)(2)(\lambda)(\lambda - 1) = 2\lambda(\lambda - 1).$$

If $\lambda \neq 0, 1$, the four diagonal entries of U_λ are nonzero. Thus there are four pivot positions, and the pivot criterion gives

$$A_\lambda \text{ is invertible for } \lambda \neq 0, 1.$$

The determinant calculation independently identifies the same exceptional parameter values, but the invertibility conclusion here follows from the pivot criterion.

Solution to Problem 6: The Case $\lambda = 0$

Substitution into the parameter-free triangular form gives

$$U_0 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix} \xrightarrow{R_4 \leftarrow R_4 + \frac{1}{3} R_3} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Hence there are three pivots and one free variable. The homogeneous equations give

$$x_4 = 0, \quad x_3 = -2x_2, \quad x_1 = -2x_2.$$

Taking $x_2 = -1$ yields

$$\boxed{\mathbf{x} = \begin{bmatrix} 2 \\ -1 \\ 2 \\ 0 \end{bmatrix}}, \quad A_0 \mathbf{x} = \mathbf{0}.$$

Thus A_0 is singular.

Solution to Problem 6: The Case $\lambda = 1$

For $\lambda = 1$,

$$U_1 = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Again there are three pivots and one free variable. The homogeneous equations give

$$x_3 = -3x_4, \quad x_2 = \frac{3}{2}x_4, \quad x_1 = -4x_4.$$

Taking $x_4 = 2$ yields

$$\mathbf{x} = \begin{bmatrix} -8 \\ 3 \\ -6 \\ 2 \end{bmatrix}, \quad A_1 \mathbf{x} = 0.$$

Thus A_1 is singular. Consequently,

A_λ is invertible exactly when $\lambda \neq 0, 1$.

Solution to Problem 6: Transformations and Linear Systems

By the Invertible Matrix Theorem,

$$\lambda \neq 0, 1 \iff T_\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}^4 \text{ is both one-to-one and onto.}$$

Hence, when $\lambda \neq 0, 1$, the equation

$$A_\lambda \mathbf{x} = \mathbf{b}$$

has exactly one solution for every $\mathbf{b} \in \mathbb{R}^4$.

When $\lambda = 0$ or $\lambda = 1$, the reduced matrix has three pivots:

- T_λ is not one-to-one because $A_\lambda \mathbf{x} = 0$ has a nonzero solution;
- T_λ is not onto because the square matrix has no pivot in one row;
- some right-hand sides give no solution;
- whenever the system is consistent, one variable is free, so there are infinitely many solutions.